

Dear Hiring Team,

I am writing to apply for the 2026 Summer Data Science Internship at Hone. I am currently pursuing a B.S. in Data Science and Big Data Technology at The Chinese University of Hong Kong, Shenzhen.

I previously worked as an algorithm engineering intern at Li Auto, where I was responsible for developing productivity tools leveraging large language models (LLMs) to automate code generation for in-house automotive software development, significantly enhancing R&D efficiency in a vertical domain. My work includes generating code with analyzing, integrating models with internal APIs, and deploying solutions on internal GPU clusters. This experience has strengthened my ability to turn ambiguous technical problems into practical solutions and improved my communication skills through technical documentation and internal training.

My research experience has further developed my analytical and modeling skills. I have engaged in research on graph theory, including the densest subgraph problem on directed graphs and the approximate maximum clique problem in bipartite graphs. Through these projects, I have learned how to approach open-ended problems rigorously and translate theory into implementation.

Hone is dedicated to improving health and extending life through science and technology, and this mission deeply inspires me. I am looking forward to applying my skills in Python, machine learning, and data processing to contribute to the company's products.

Thank you for your time and consideration. I would welcome the opportunity to discuss my background further.

Sincerely,

Zexuan Kang

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EDUCATION BACKGROUND

The Chinese University of Hong Kong, Shenzhen(CUHKSZ), China 09/2022 - Expected. 06/2026
Bachelor of Science in Data Science and Big Data Technology | GPA: 3.4/4.0

Core Coursework: Introduction to Data Science, Data Structures, Optimization, Machine Learning, Programming Methodology(Python/C/C++), Design and Analysis of Algorithms, Cloud Computing, Introduction to Quantum Computation and Quantum Information, Introduction to Coding Theory, Calculus I&II, Linear Algebra, Probability, Discrete Mathematics, Stochastic Processes, Stochastic Simulation, Investment Analysis and Portfolio Management.

WORK EXPERIENCES

Li Auto Inc., *Algorithm Engineer Intern, China* 06/2025 -08/2025

- Developed productivity tools leveraging large language models (LLMs) to automate code generation for in-house automotive software development, significantly enhancing R&D efficiency in a vertical domain.
- Conducted domain-specific code demand analysis and independently integrated LLMs with automotive software APIs, achieving 95% accuracy in code outputs and deploying solutions on internal GPU clusters.
- Coordinated across research and development teams for data annotation and agile development, ensuring scalable and reliable deployment of LLM-based tools; authored internal technical documentation and led seminars and training sessions to promote adoption and responsible use of LLM applications across engineering departments.

Shanghai Baoshu Tech Co., Ltd., *Product R&D Intern, China* 05/2023 -08/2023

- Contributed to a core robotics R&D project, programmed the intelligent teaching robot movements, and designed the demonstration workflow, ensuring smooth deployment in real-world applications.
- Built and applied a proprietary corpus for "Hualong N.O. 1" (a nuclear power unit); responded to data annotation and algorithm testing; incorporated deep learning using TensorFlow and PyTorch to fine-tune model performance, achieving notable improvements in accuracy and efficiency.

RESEARCH EXPERIENCE

Anchored Densest Subgraph Problem (ADSS) on Directed Graphs 09/2024 – 01/2025

- Identified a gap in conventional densest subgraph research where key anchor structures are often overlooked. Initiated a study on the Anchored Densest Subgraph (ADSS) problem, which aims to find a subgraph with the highest average edge density while containing a predefined target subgraph.
- Derived a dynamic programming formulation from the linear programming model and applied the Frank-Wolfe algorithm to optimize the solution under structural constraints. Implemented the algorithm and developed a quality evaluation framework to validate its effectiveness on real-directed graph data.
- Preliminary results demonstrate that the proposed approach effectively balances density maximization and constraint satisfaction with significant improvements in computational efficiency.

Approximate Maximum Clique Problem in Bipartite Graphs 09/2023- 05/2024

- Conducted research on the defective-clique problem in bipartite graphs, inspired by gaps found in existing literature on network clustering.
- Independently developed and optimized algorithms using alpha-beta core compression and ordering techniques. These optimizations reduced computation time by 10%, significantly accelerating the clique-finding process and offering practical insight into solving defective clique problems in bipartite settings.

Beijing Mobile Customer-Experience Analytics, *MathorCup* 01/2023

- Demonstrated the theoretical equivalence between Pearson correlation and Euclidean distance on standardized data, then quantified links between 50+ network quality variables and overall satisfaction, isolating the top five drivers for both voice and mobile data services.
- Built and tuned XGBoost with feature crossing and regularization; achieved 85% prediction accuracy on potential customer rating sets and auto-generated result sheets required by the competition platform.

SKILLS & CERTIFICATES & HONORS

Programming Languages & Tools: Python, C++, TensorFlow, PyTorch

Algorithmic Proficiency: Graph Theory, Network Flow, Dynamic Programming, Recommender Systems

Honors & Awards: Bronze Medal, ACM-ICPC Regional Contest | First Prize, CUHK(SZ) Algorithm Programming Competition | Second Prize, National Olympiad in Informatics in Provinces (NOIP)